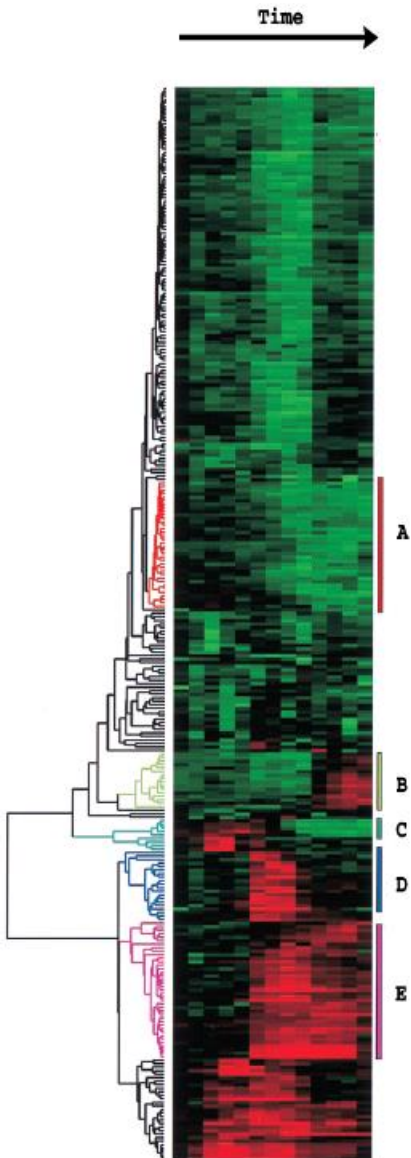


R

鍾翊方(R308, 67358)
ifchung@nycu.edu.tw

Array Data



Multi-Samples/Conditions Expression Analysis

gene \ sample	A	B	C	...	...	...
1	0.6	0.4	0.2	...		
2	0.2	0.9	0.8	...		
3	0	0	0.3	...		
4	0.7	0.5	0.2	...		
..	..	..	..	...		
1000	0.3	0.8	0.7	...		

Learning set

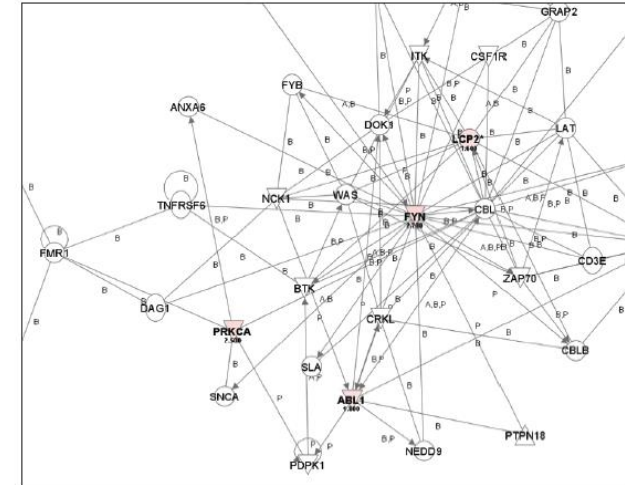
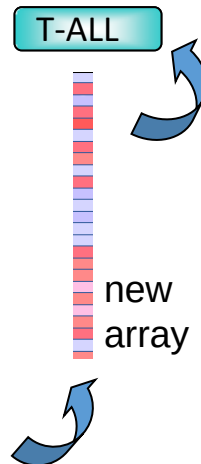
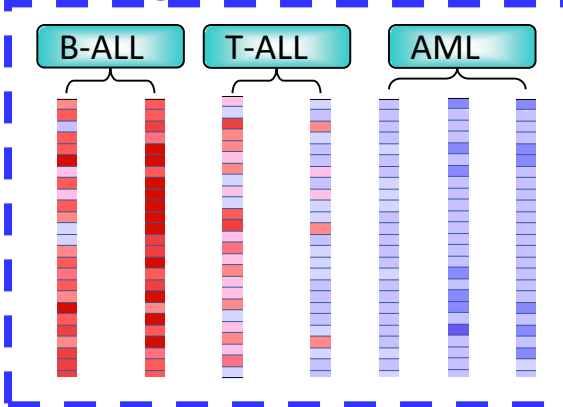


Figure 3: A Network Representing the BCR-ABL Subtype.

Data Set (SRBCT)

- Small Round Blue Cell Tumor (SRBCT) data set
 - Original paper reported by *Khan et al.* (Nature, 2001)
 - Well studied by many different research groups
 - A cDNA microarray data set with 88 samples for 4 different classes
 - 2308 genes were used

	Total 88 samples		
	Original set	Indep. set	Non-SRBCT
	63 samples	20 samples	5 samples
Ewing family of tumors (EWS)	23	6	--
Burkitt lymphomas (BL)	8	3	--
Neuroblastoma (NB)	12	6	--
Rhabdomyosarcoma (RMS)	20	5	--
Non-SRBCT	--	--	5

Data Set (Leukemia)

- Leukemia data set
 - Original paper reported by *Armstrong et al.* (*Nature Genetics*, 2002)
 - Well studied by many different research groups
 - An Affymetrix oligonucleotide microarray data set which contains **72 samples** with **3** different classes
 - **12,582 genes** were used

	Original set 57 samples	Indep. set 15 samples
Acute lymphoblastic leukemia (ALL)	20	4
Mixed-lineage leukemia (MLL)	17	3
Acute myelogenous leukemia (AML)	20	8

R統計軟體

- R是什麼, 為什麼要用 R ?
- 安裝與設定
- 基本操作
- 安裝套件package
- 基本繪圖
- Demo

R是什麼

- R 系統是由 Ross Ihaka 與 Robert Gentleman 從 S 語言所發展出來, 主要是為了統計分析與繪圖
- S 語言早在 1980 年代末期, 由 AT&T 實驗室, Rick Becker, John Chambers, 與 Allan Wilks 發展用來進行統計分析與作圖
- Insightful 公司將 S 商品化, 並加入許多方便的操作介面, 稱為 S-plus
- R 可視為統計數學軟體, 也是一種程式語言

R是什麼

- R是一個開放原始碼 (open-source, GNU General Public License)的統計分析軟體
- R跟S都是以物件導向為主的程式語言
- R的網站 <http://www.r-project.org>
(在Google搜尋R就可以找到)
- R最新版本為3.32 (released on Monday 2016-10-31)

為什麼要用 R ？

- R 與 S-plus 語法大多相近,但是 R 是開放原始碼 (open-source, GNU General Public License)(免費,自由軟體), 目前由一群跨國際的志工人員組成的 R 核心發展組織 (R core-development team) 所維持, 運作與持續更新.
- R 與 S 都是以物件導向為主的程式語言, 透過交互作用方式很容易地進行統計分析與繪圖,這與 SAS, SPSS 的方式有所不同

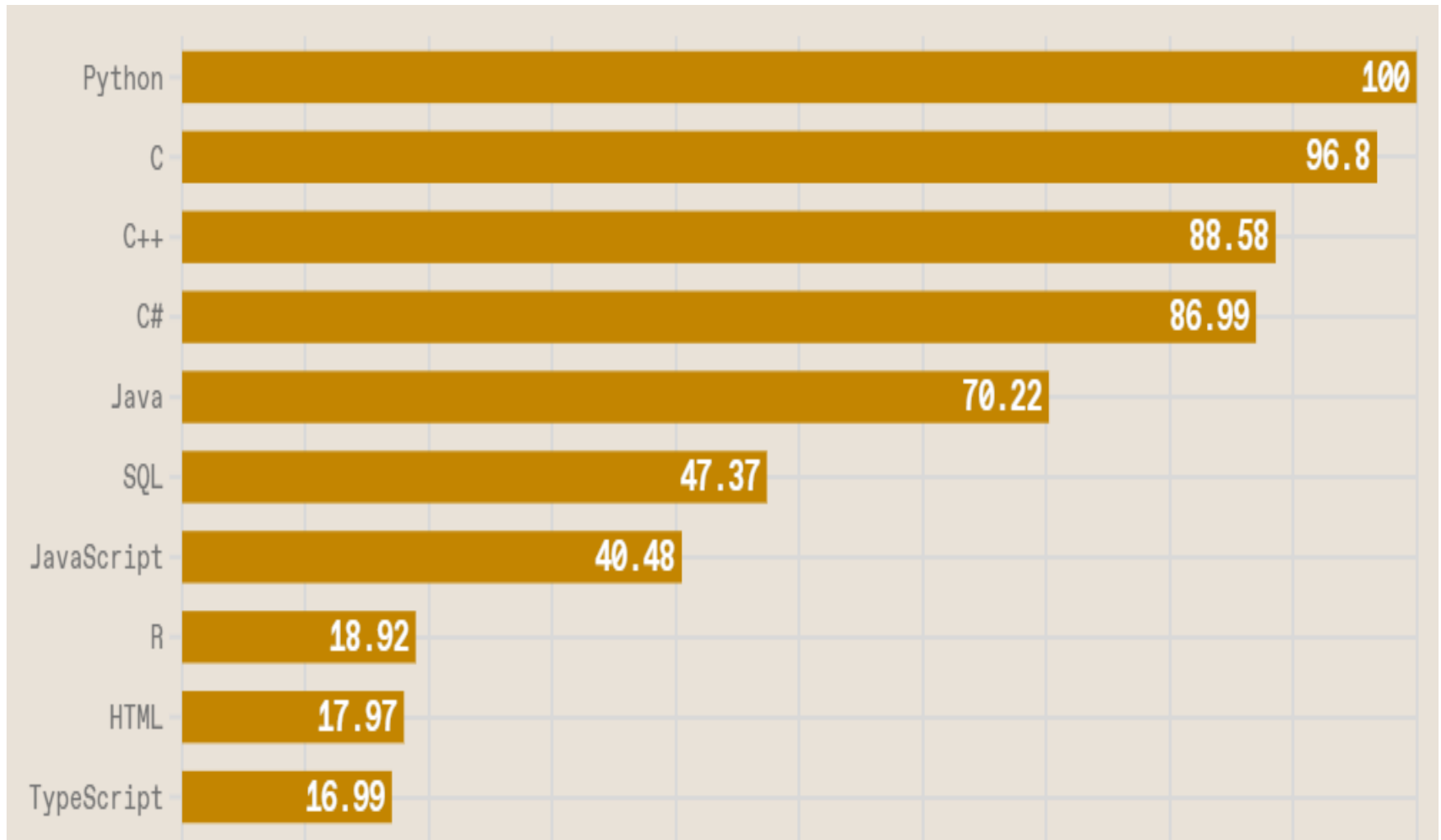
為什麼要用 R ？

- R 是科學家寫出來的軟體,而且被科學家所使用
- 可以在不同平台上執行 Windows, Mac, Unix, Linux, ...
- 簡單安裝
- R 是免錢的！
- 有數百個擴充套件(**packages**)可以安裝
- 有許多文章的發表都順帶包含了 R 的套件

為什麼要用 R ?

- 有很棒的線上說明及文件可以參考
- 還有郵件論壇可以發問(mailing list)
- 很棒的社群(一堆統計專家?) – friendly and helpful people...
- 很容易可以呼叫其他語言的函式庫
Fortran, C, Java, Python... libraries

The Top Programming Languages 2022



From IEEE Spectrum

<https://spectrum.ieee.org/top-programming-languages-2022>

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R軟體：應用統計方法 修訂版 附光碟1片

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出版社：東華 [訂閱出版社新書快訊](#) 

出版日期：2014/01/01

語言：繁體中文

定價：720元

優惠價：95折 684元



【分級買就送】分級會員買就送OPENPOINT(部份除外) [詳情](#)

運送方式：

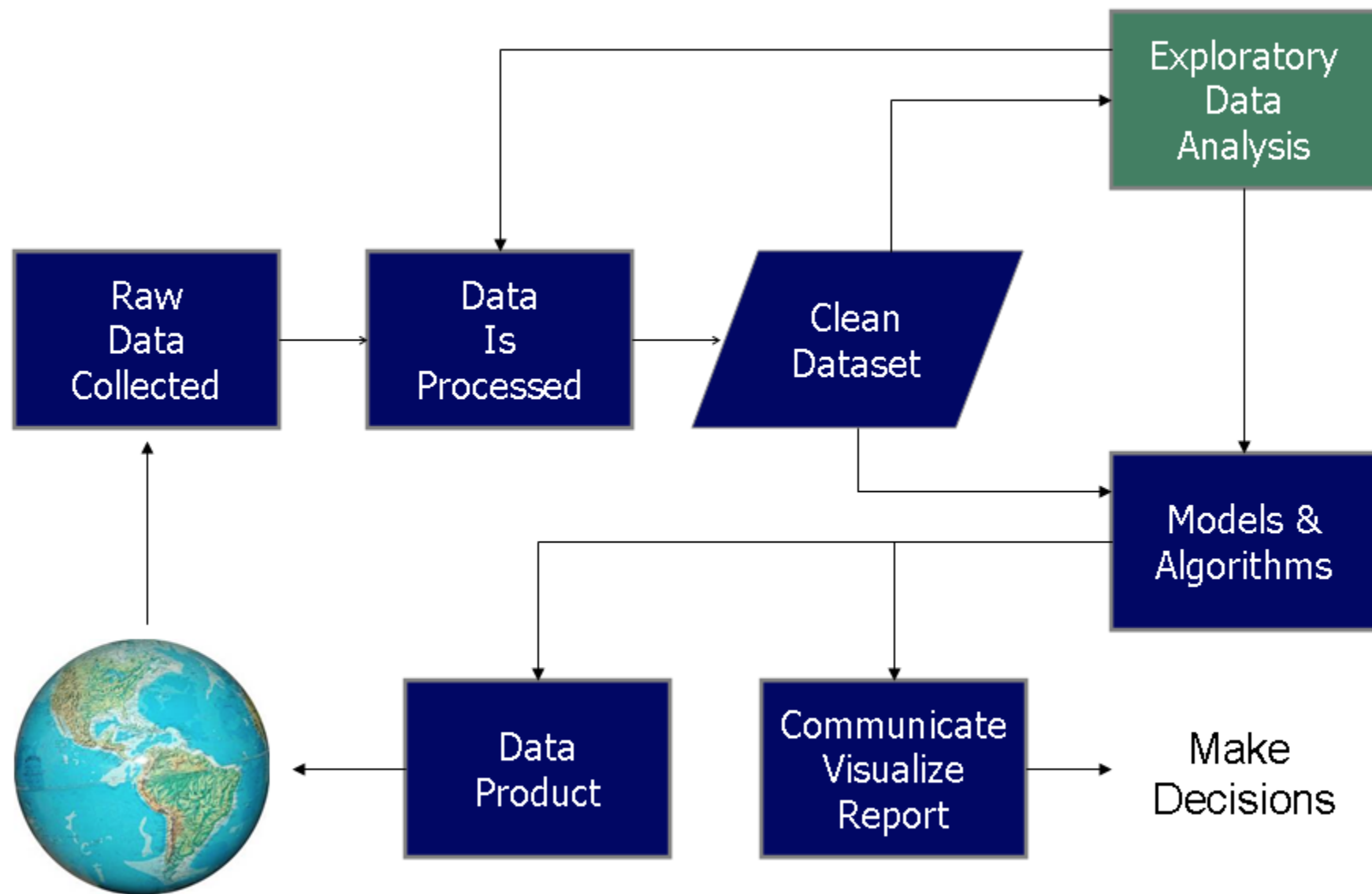


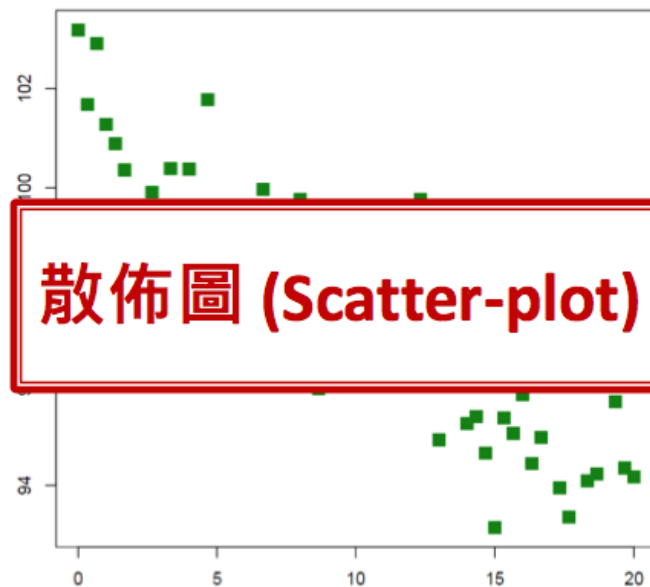
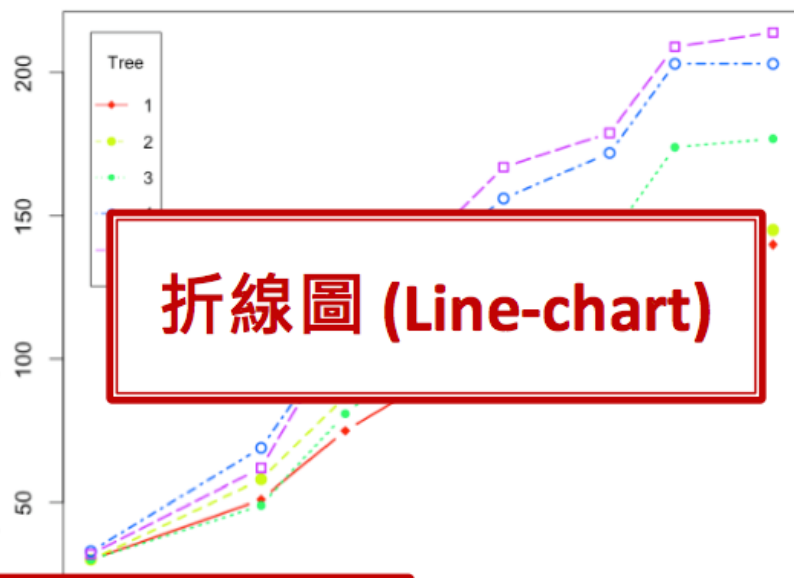
可配送點：台灣、蘭嶼、綠島、澎湖、金門、馬祖、全球



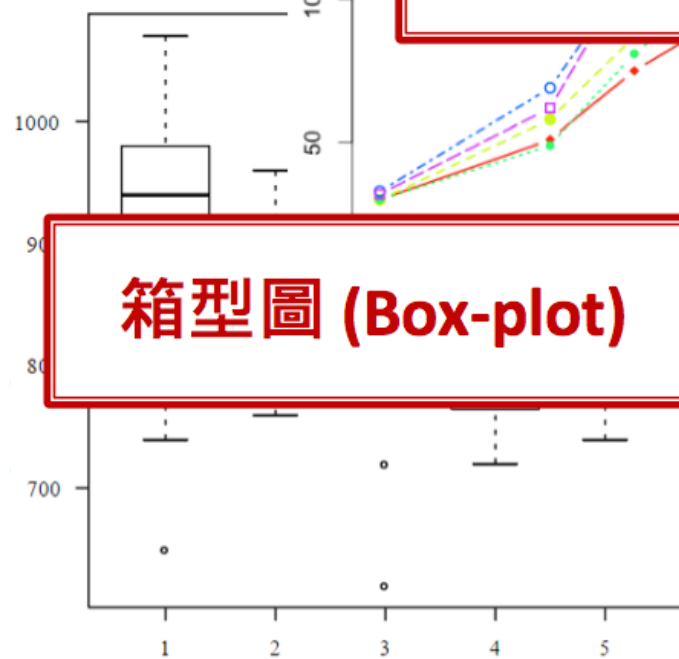
可取貨點：台灣、蘭嶼、綠島、澎湖、金門、馬祖
香港、澳門、新加坡

Data Science Process



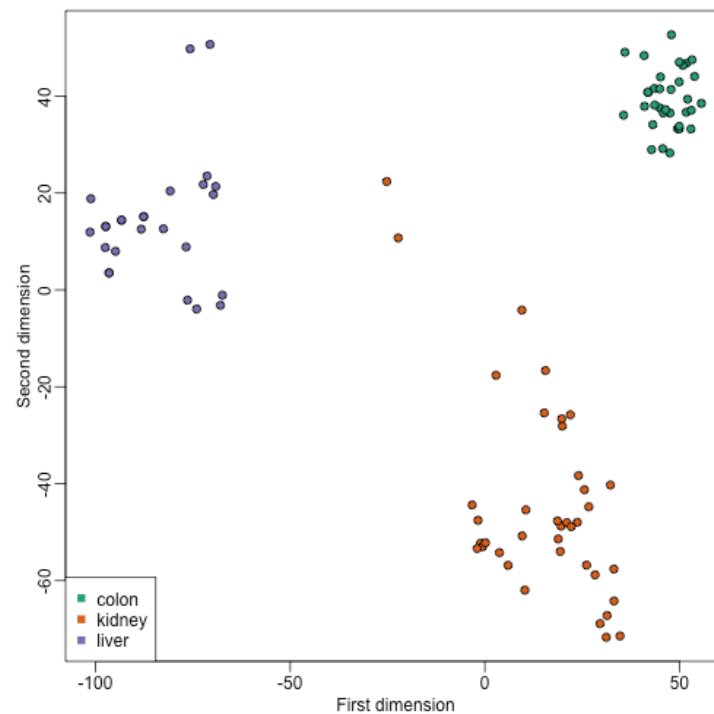
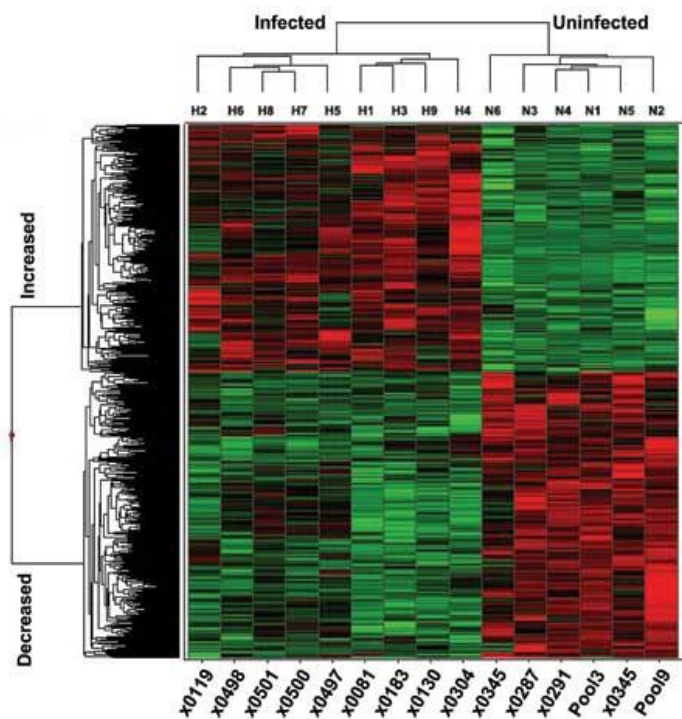


箱型圖 (Box-plot)

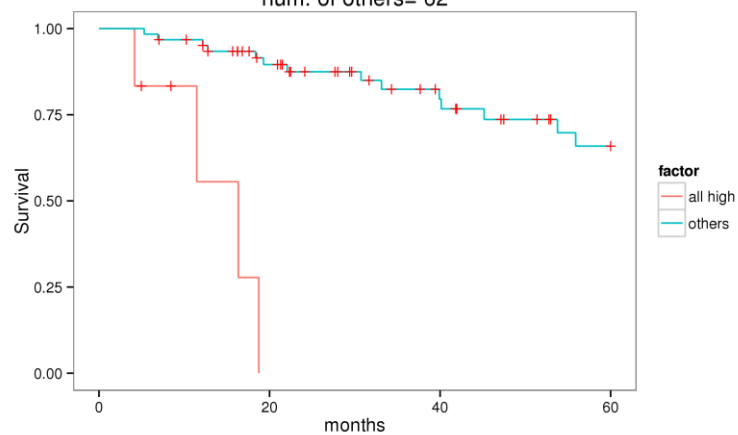


直方圖 (Histogram)

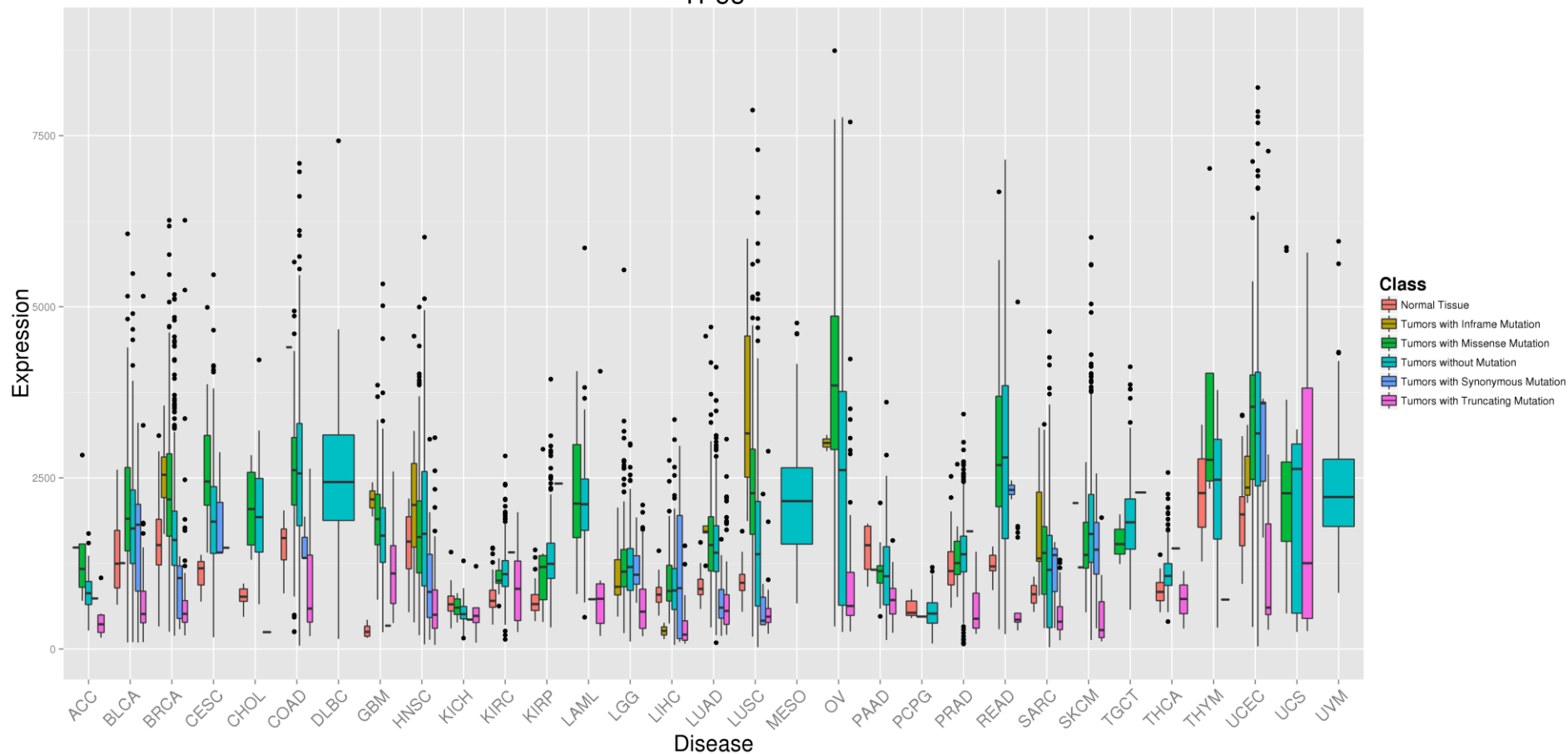




Adrenocortical_carcinoma (N-)
P value=6.55e-09
num. of all high= 6
num. of others= 62



TP53



R & Rstudio

-安裝與使用

安裝與設定

- 下載R <http://www.r-project.org>
- 安裝R
- 下載RStudio <https://www.rstudio.com/>
- 安裝RStudio
- 設定RStudio

<http://www.r-project.org/>



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The R Project for Statistical Computing

Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To **download R**, please choose your preferred [CRAN mirror](#).

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- **R version 4.4.1 (Race for Your Life)** has been released on 2024-06-14.
- We are deeply sorry to announce that our friend and colleague Friedrich (Fritz) Leisch has died. [Read our tribute to Fritz here](#).
- **R version 4.4.0 (Puppy Cup)** has been released on 2024-04-24.
- **R version 4.3.3 (Angel Food Cake)** (wrap-up of 4.3.x) was released on 2024-02-29.

Sweden

<https://ftp.acc.umu.se/mirror/CRAN/>

Switzerland

<https://stat.ethz.ch/CRAN/>

<https://mirror.metanet.ch/cran/>

Taiwan

<https://cran.csie.ntu.edu.tw/>

Thailand

<http://mirrors.psu.ac.th/pub/cran/>

Academic Computer Club, Umeå University

ETH Zürich

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National Taiwan University, Taipei

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Taiwan mirror



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The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#) (Debian, Fedora/Redhat, Ubuntu)
- [Download R for macOS](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want the precompiled binaries in the upper box, not the source code. The sources have to be compiled before you can use them. If you want to do it!

選擇Windows

- The latest release (2023-06-16, Beagle Scouts) [R-4.3.1.tar.gz](#), read [what's new](#) in the latest version.
- Sources of [R alpha and beta releases](#) (daily snapshots, created only in time periods before a planned release).
- Daily snapshots of current patched and development versions are [available here](#). Please read about [new features and bug fixes](#) before filing corresponding feature requests or bug reports.
- Source code of older versions of R is [available here](#).
- Contributed extension [packages](#)

Questions About R

- If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.



R for Windows

Subdirectories:

[base](#)

Binaries for base distribution. This is what you want to [install R for the first time](#).

[contrib](#)

Binaries of contributed CRAN packages (for R $\geq$ 2.13.x; managed by Uwe Ligges). There is also information on [third party software](#) available for CRAN Windows services and corresponding environment and make variables.

[old contrib](#)

Binaries of contributed CRAN packages for outdated versions of R (for R $<$ 2.13.x; managed by Uwe Ligges).

[Rtools](#)

Tools to build R and R packages. This is what you want to build your own packages on Windows, or to build R itself.

Please do not submit binaries for Windows binaries.

請選**base**

Uwe Ligges directly in case of questions / suggestions related to

You may also want to read

contrib是套件庫

Note: CRAN does some checks, but we cannot guarantee. Use the normal precautions with downloaded executables.

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R-4.4.1 for Windows

[Download R-4.4.1 for Windows](#) (82 megabytes, 64 bit)

[README on the Windows binary distribution](#)

[New features in this version](#)

This build requires UCRT, which is part of Windows since Windows 10 and Windows Server 2016. On older systems, UCRT has to be installed manually from [here](#).

If you want to double-check that the package you have downloaded matches the package distributed by CRAN, you can compare the [md5sum](#) of the .exe to the [fingerprint](#) on the master server.

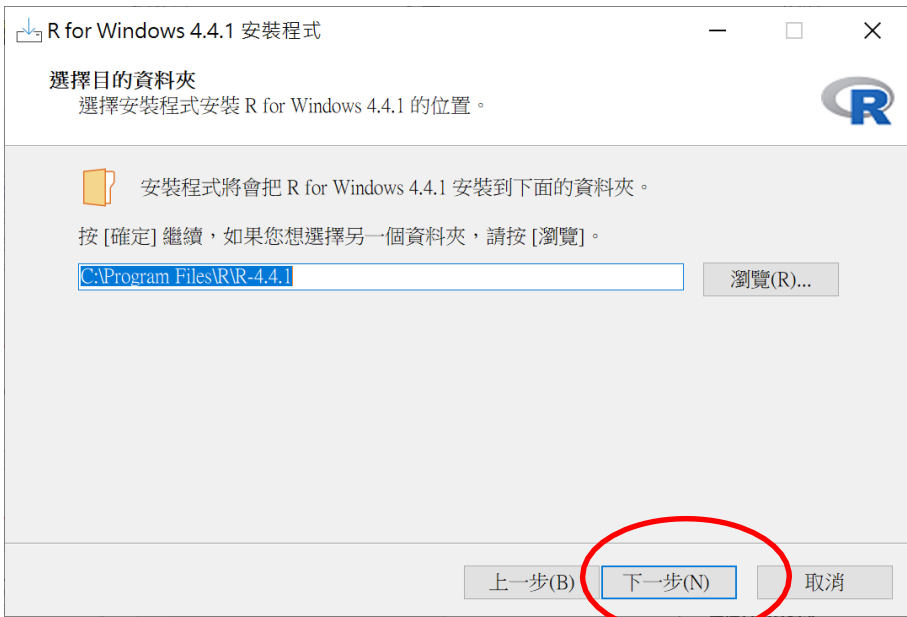
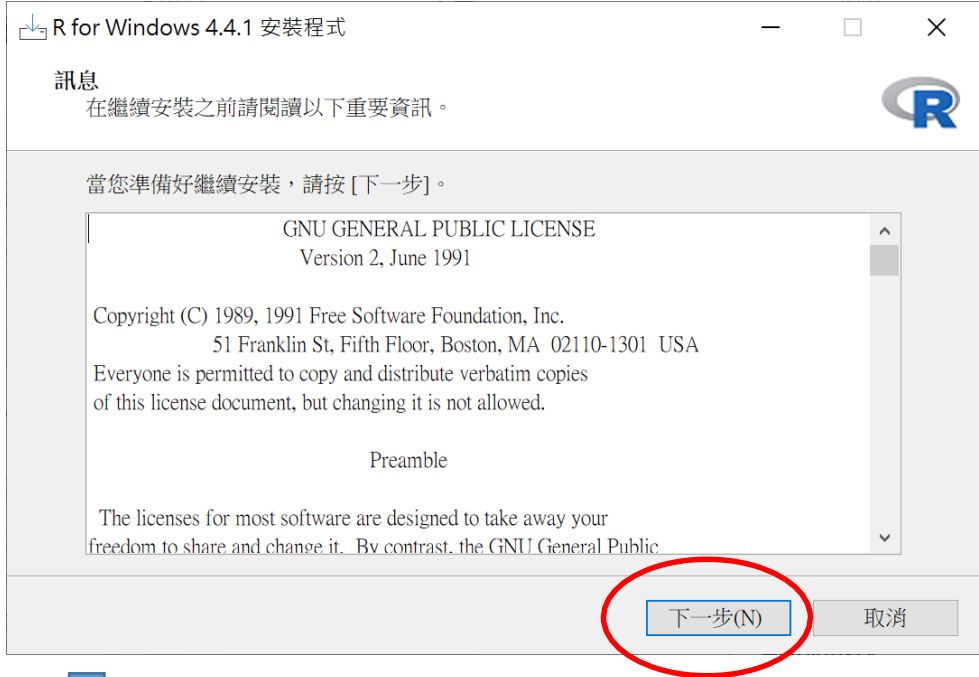
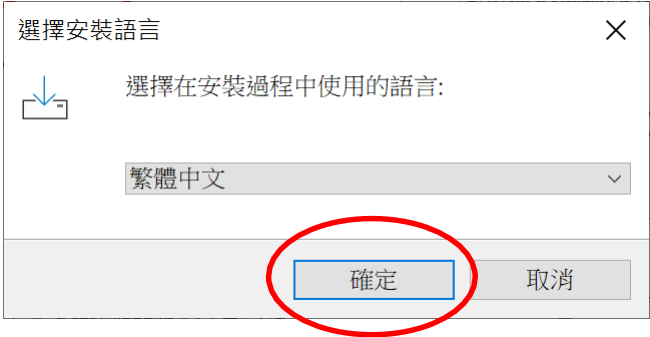
Frequently asked questions

- [Does R run under my version of Windows?](#)
- [How do I update packages in my previous version of R?](#)

Please see the [R FAQ](#) for general information about R and the [R Windows FAQ](#) for Windows-specific information.

Other builds

- Patches to this release are incorporated in the [r-patched snapshot build](#).
- A build of the development version (which will eventually become the next major release of R) is available in the [r-devel snapshot build](#).
- [Previous releases](#)



安裝的檔案夾位置



R Console

```
R version 4.4.1 (2024-06-14 ucrt) -- "Race for Your Life"  
Copyright (C) 2024 The R Foundation for Statistical Computing  
Platform: x86_64-w64-mingw32/x64
```

R 是自由軟體，不提供任何擔保。
在某些條件下歡迎您將其散佈。
用 `'license()'` 或 `'licence()'` 來獲得散佈的詳細條件。

R 是個協作計劃，有許多人為之做出了貢獻。
用 `'contributors()'` 來看詳細的情況以及
用 `'citation()'` 會告訴您如何在出版品中正確地參照 R 或 R 套件。

用 `'demo()'` 來看一些示範程式，用 `'help()'` 來檢視線上輔助檔案，或
用 `'help.start()'` 透過 HTML 瀏覽器來看輔助檔案。
用 `'q()'` 離開 R。

```
> |
```

為什麼要裝RStudio？

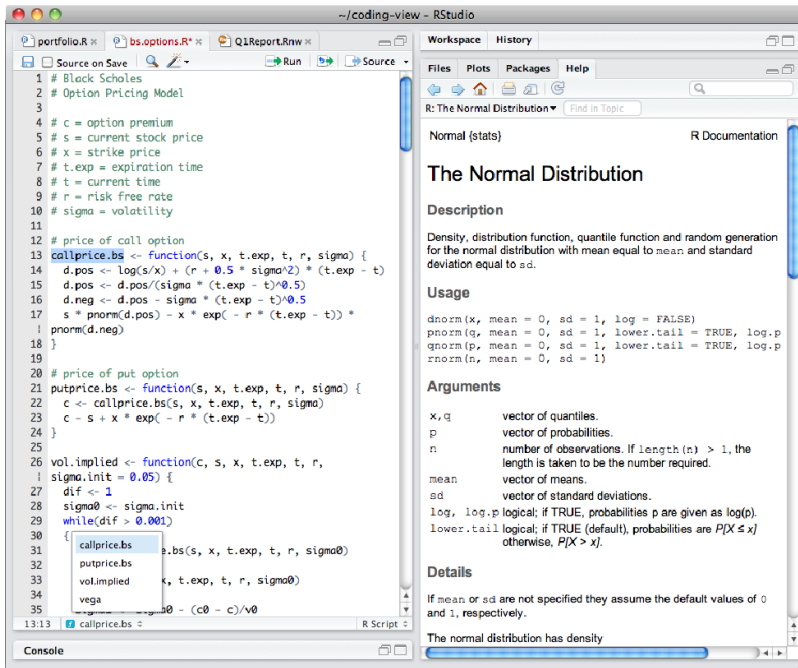
- 什麼是RStudio？
 - 它是R的IDE，(整合開發環境，Integrated Development Environment的英文縮寫，可輔助開發程式的應用軟體。From Wikipedia)
- 為什麼要用RStudio？
 - 支援語法高亮Syntax highlighting
 - 支援代碼自動完成code completion
- – 工作空間管理與計畫檔案管理
 - – 免費、開放原始碼、支援不同作業系統平台 (windows、Mac、Linux)、支援版本控制系統(Git、Subversion)

語法高亮 Syntax highlighting

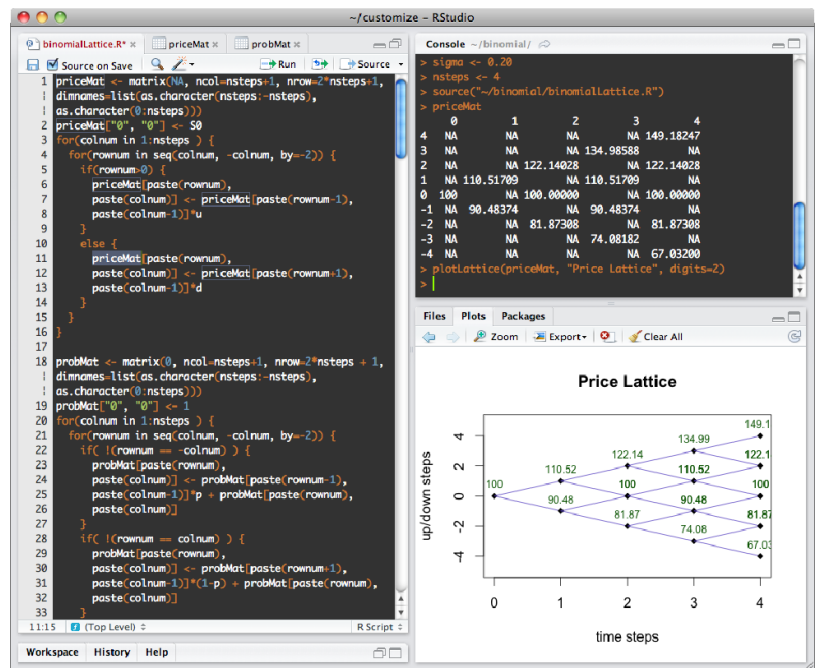
```
1 C#####
2 C FOR NGA Data *.AT2 only
3 C Only one component
4 C Read NGA ACC data get PGA、PGV、PGD into a list
5 C
6 C#####
7 $debug
8 PROGRAM Rd_NGA_PGA
9
10 PARAMETER(maxntp=100000)
11 INTEGER ncs(3)
12 CHARACTER args(3)*128, buff*100, rfile*128
13 CHARACTER SGM_ID*17, STA_ID*6
14 INTEGER npts
15 REAL dt, PGA, PGV, PGD, dm, tmd, vm, tmv, amax, tma
16 REAL acc(maxntp), time(maxntp), a(maxntp)
17
18 Call GETARG(1,args(1),ncs(1))
19 Call GETARG(2,args(2),ncs(2))
20 Call GETARG(3,args(3),ncs(3))
21
22 SGM_ID = TRIM(args(1))
23 STA_ID = TRIM(args(2))
24 rfile = TRIM(args(3))
25
26 PRINT *,SGM_ID
27 PRINT *,STA_ID
28
29 OPEN(1, file=rfile)
30 DO I=1,3
31 READ(1,'(A100)') buff
32 ENDDO
33 READ(1,*) npts,dt
34 READ(1,*) (acc(J),J=1,npts)
35 CLOSE(1)
36
37 kg=npts
38 DO I=1,npts
39 time(i)=(i-1)*dt
40 ENDDO
41
42 C
43 C...find max velocity, displacement, and acceleration
44 C
45 a=acc
46 amax=0.0
```

Ln 22, Col 29, C0 DOS Mod: 2012/5/2 下午 06 File Size: 2282 INS CAP

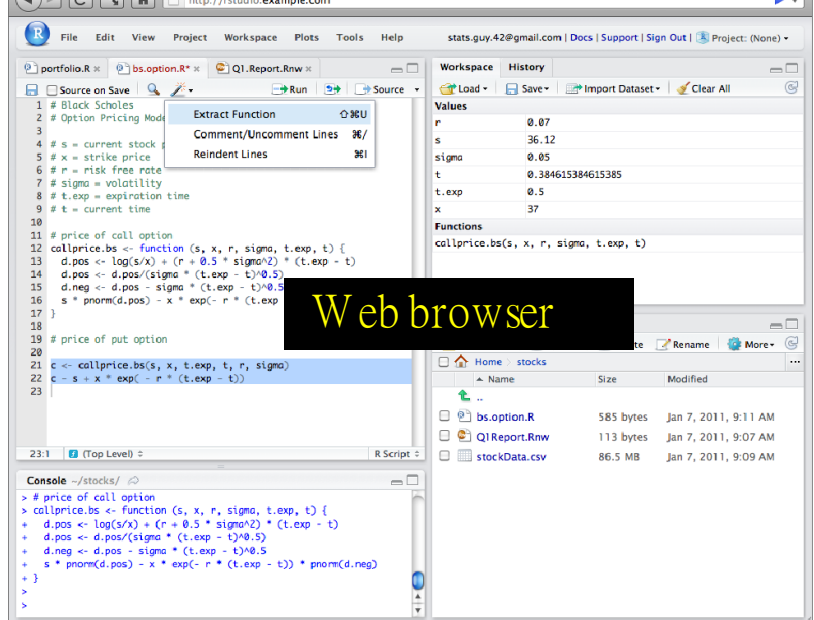
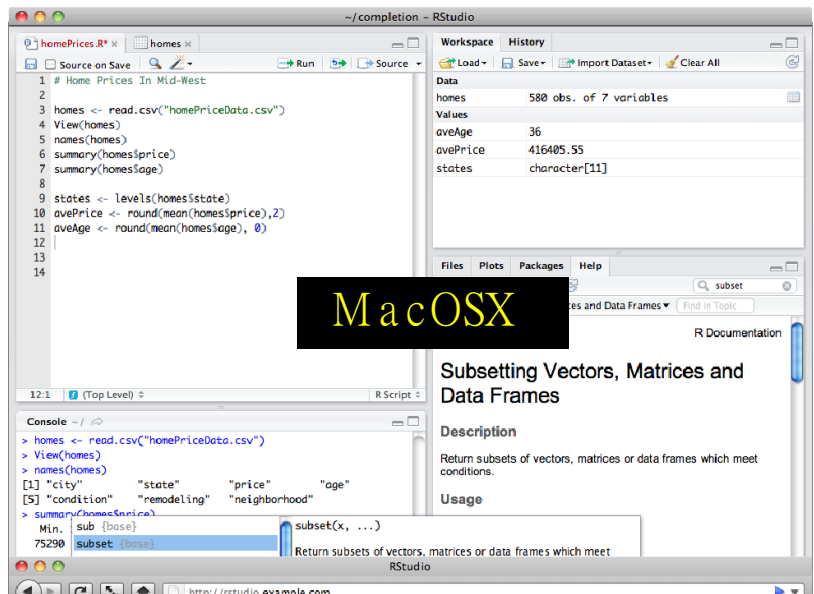
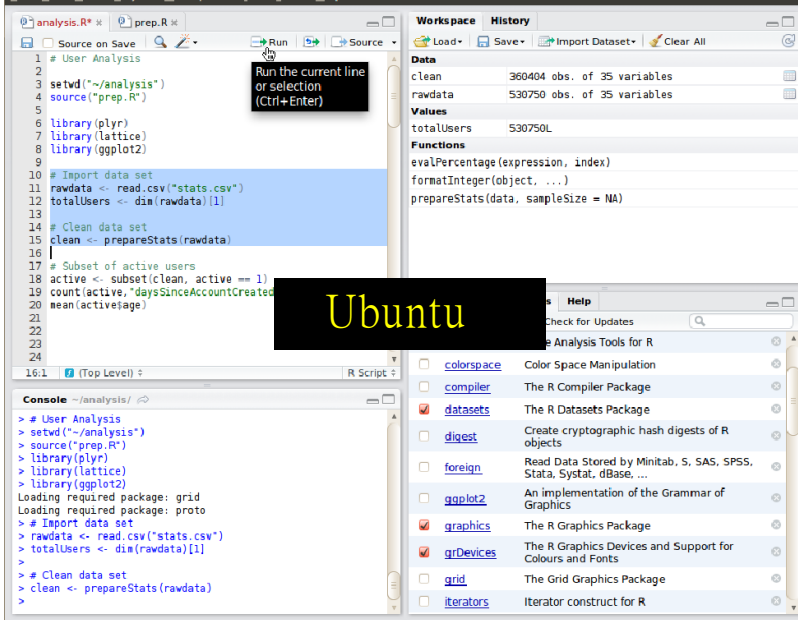
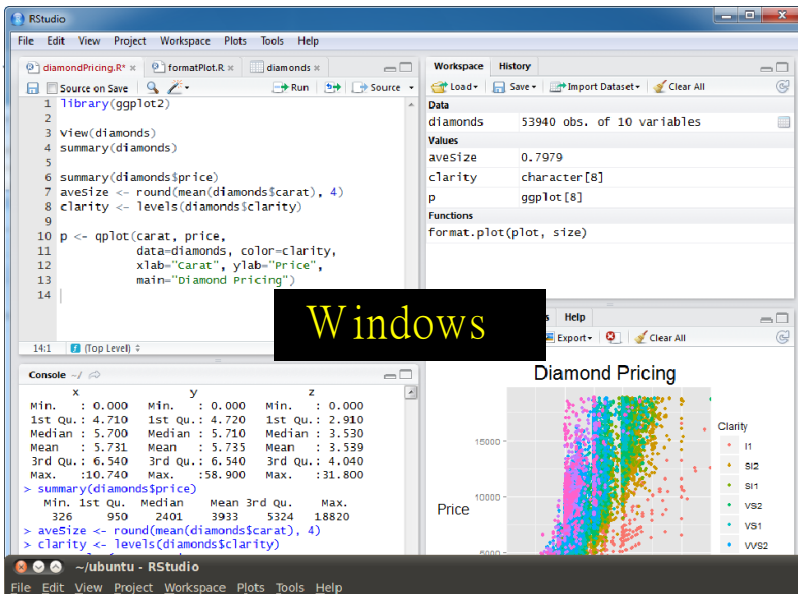
個人化配置



自定版面配置



自定版面主題配色



```
1 # User Trend Analysis
2 # Breakdown of active and non-active users
3
4 library(plyr)
5 library(ggplot2)
6
7 userData <- read.csv("userDataTrends.csv")
8 userData <- subset(userData, select = -c(id, group))
9 userData$active <- as.factor(userData[,1])
10
11 states <- levels(userData$state)
12
13 names(userData)
14 count(userData, "active")
15 View(userData)
16
17 summary(subset(userData, active == 1)$state)
18 summary(subset(userData, active == 0)$state)
19
20 qplot(state, age, color = active, data = userData,
21       main = "Breakdown of Users by Age and State") +
22       opts(plot.title = theme_text(size = 19))
23
```

程式碼編輯區

```
Console ~/
active...1 freq
1 FALSE 310
2 TRUE 270
> View(userData)
> summary(subset(userData, active == 1)$state)
IA IL IN KS MI MN MO ND NE OH SD
19 26 21 21 27 49 22 26 19 16 24
> summary(subset(userData, active == 0)$state)
IA IL IN KS MI MN MO ND NE OH SD
26 27 18 31 27 49 22 32 19 33 26
> qplot(state, age, color = active, data = userData,
+       main = "Breakdown of Users by Age and State") +
+       opts(plot.title = theme_text(size = 19))
>
```

程式執行區

Workspace History

Load Save Import Dataset Clear All

Data

userData 580 obs. of 5 variables

Values

active integer[270]

states character[11]

Functions

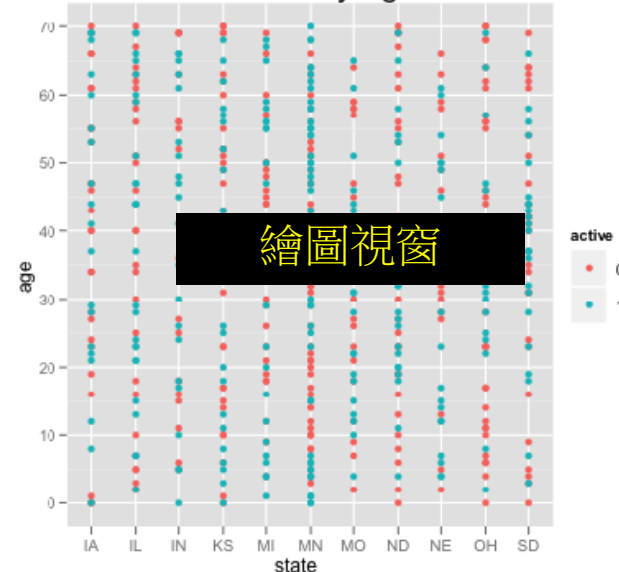
split(group, location, ...)

記憶體中的物件管理

Files Plots Packages Help

Zoom Export Clear All

Breakdown of Users by Age and State



繪圖視窗

~/history - RStudio

Workspace History

Search results: qplot

qplot(carat, price, data=diamonds) 2011-03-03 07:12

qplot(carat, price, data=diamonds, color=clarity, xlab="Carat", ylab="Price", main="Diamond Pricing") 2011-03-02 14:03

+ coord_cartesian(xlim=c(0, 3.5)) + opts(plot.title=theme_text(size=23))

qplot(carat, binwidth=0.05, data=diamonds) 2011-03-02 14:00

qplot(log(x), log(y)) + geom_smooth(method="lm") 2011-03-01 13:23

qplot(x, y, xlab="Year", ylab="Users in Mi") 2011-03-01 13:20

qplot(x, y) + geom_smooth(method="loess") 2011-03-01 13:18

qplot(x, y) 2011-03-01 13:17

Files Plots Packages Help

Zoom Export Clear All

Diamond Pricing

Price

Carat

clarity

- I1
- S2
- S1
- VS2
- VS1
- VVS2
- VVS1
- IF

Source on Save Run Source

```
1 library(ggplot2)
2
3 View(diamonds)
4 summary(diamonds)
5 summary(diamonds$price)
6
7 qplot(carat, price, data=diamonds)
8
9 qplot(carat, price, data=diamonds, color=clarity,
10       xlab="Carat", ylab="Price",
11       main="Diamond Pricing") +
12       coord_cartesian(xlim=c(0, 3.5)) +
13       opts(plot.title=theme_text(size=23))
14
15
```

14:1 (Top Level) R Script

Console ~

x	y	z
Min. : 0.000	Min. : 0.000	Min. : 0.000
1st Qu.: 4.710	1st Qu.: 4.720	1st Qu.: 2.910
Median : 5.700	Median : 5.710	Median : 3.530
Mean : 5.731	Mean : 5.735	Mean : 3.539
3rd Qu.: 6.540	3rd Qu.: 6.540	3rd Qu.: 4.040
Max. : 10.740	Max. : 58.900	Max. : 31.800

```
> summary(diamonds$price)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   326    950    2401    3933    5324   18820

> qplot(carat, price, data=diamonds)
> qplot(carat, price, data=diamonds, color=clarity,
+       xlab="Carat", ylab="Price", main="Diamond Pricing") +
+       coord_cartesian(xlim=c(0, 3.5)) +
+       opts(plot.title=theme_text(size=23))
>
```

~/completion - RStudio

homePrices.R* x homes x

Source on Save Run Source

```

1 # Home Prices In Mid-West
2
3 homes <- read.csv("homePriceData.csv")
4 View(homes)
5 names(homes)
6 summary(homes$price)
7 summary(homes$age)
8
9 states <- levels(homes$state)
10 avePrice <- round(mean(homes$price),2)
11 aveAge <- round(mean(homes$age), 0)
12
13
14

```

Workspace History

Load Save Import Dataset Clear All

Data

homes	580 obs. of 7 variables
Values	
aveAge	36
avePrice	416405.55
states	character[11]

Files Plots Packages Help

subset

R: Subsetting Vectors, Matrices and Data Frames Find in Topic

subset {base} R Documentation

Subsetting Vectors, Matrices and Data Frames

Description

Return subsets of vectors, matrices or data frames which meet conditions.

Usage

```
subset(x, ...)
```

Return subsets of vectors, matrices or data frames which meet conditions.

subset(x, subset, select, drop = FALSE, ...)

Press F1 for additional help

Console ~/

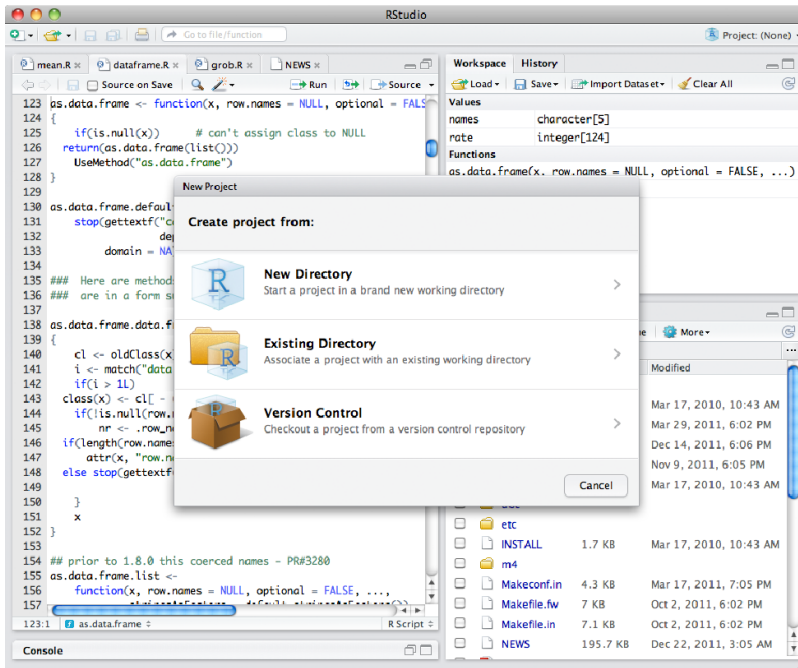
```

> homes <- read.csv("homePriceData.csv")
> View(homes)
> names(homes)
[1] "city"      "state"     "price"     "age"
[5] "condition" "remodeling" "neighborhood"
> summary(homes$price)
  Min.   sub {base}
75290 subset {base}
> summary(homes$age)
  Min.   subset.data.frame {base}
 0.00 subset.default {base}
> states subset.matrix {base}
> avePrice substitute {base}
> aveAge substituteDirect {methods}
> old <- sub

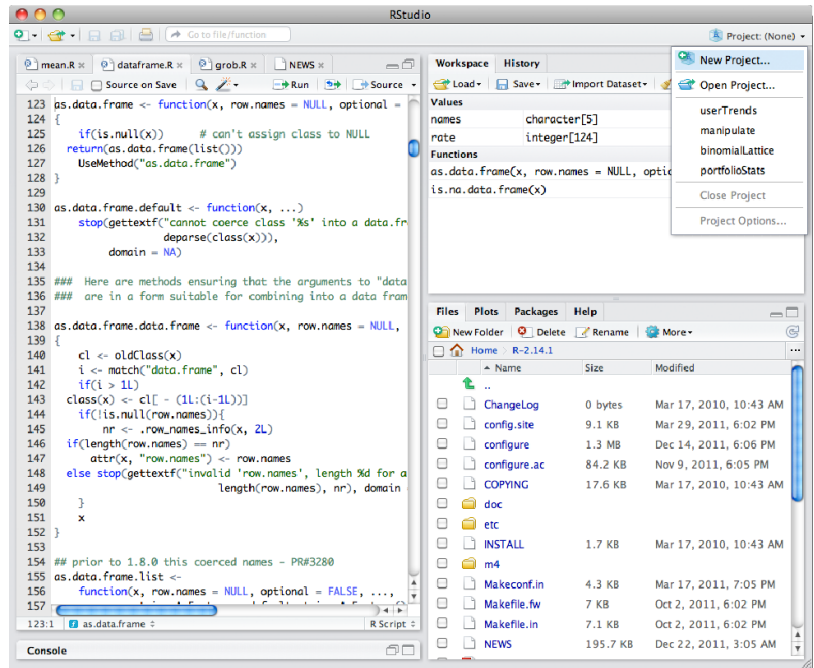
```

程式碼自動完成

計畫管理



新增計畫



於計畫之間切換

可以開始用**R**寫程式啦!!

基本操作

- `help.start()` 線上說明
- `help(mean)` 指令說明
`?mean`
- `example(plot)` 範例展示
- `data()` 顯示目前可用的資料
- `objects()`, `ls()` 顯示目前所有的物件
- `rm()` 移除物件
- 注意！！R 會區分大小寫不同！！

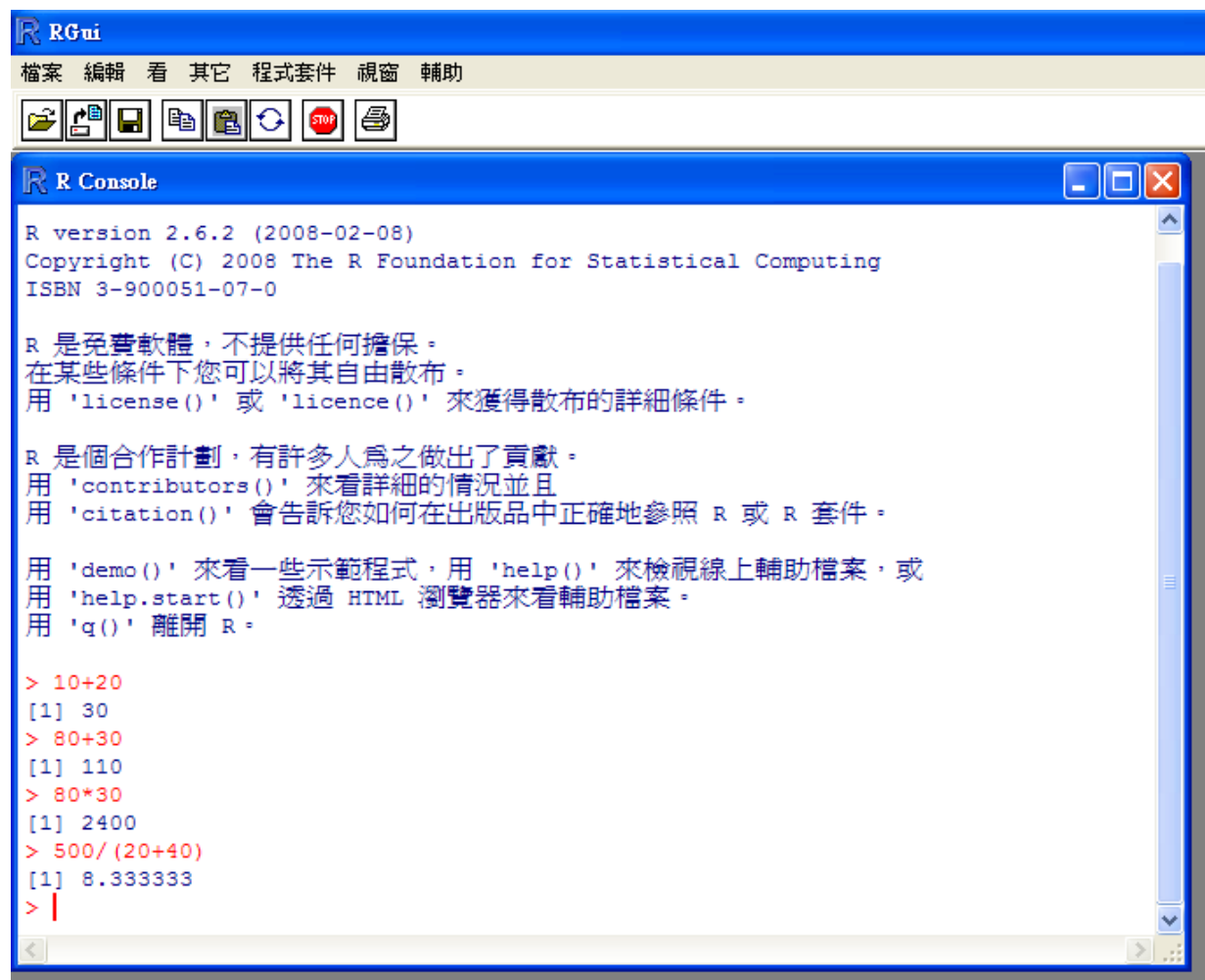
變數的分類

- "numeric", 實數向量
- "integer", 整數向量 (有時需特別指定)
- "logical", 邏輯變數向量 (true or false), 以 TRUE (T) 或 FALSE (F) 呈現,(也可以是 1 (T) 與 0 (F)).
- "complex", 複數向量
- "character", 文字或字串向量, 通常輸入時, 在文字或字串兩側加上雙引號.
- "list", 列表, 是一個由 R (S) 物件所組成的向量

R的物件類別

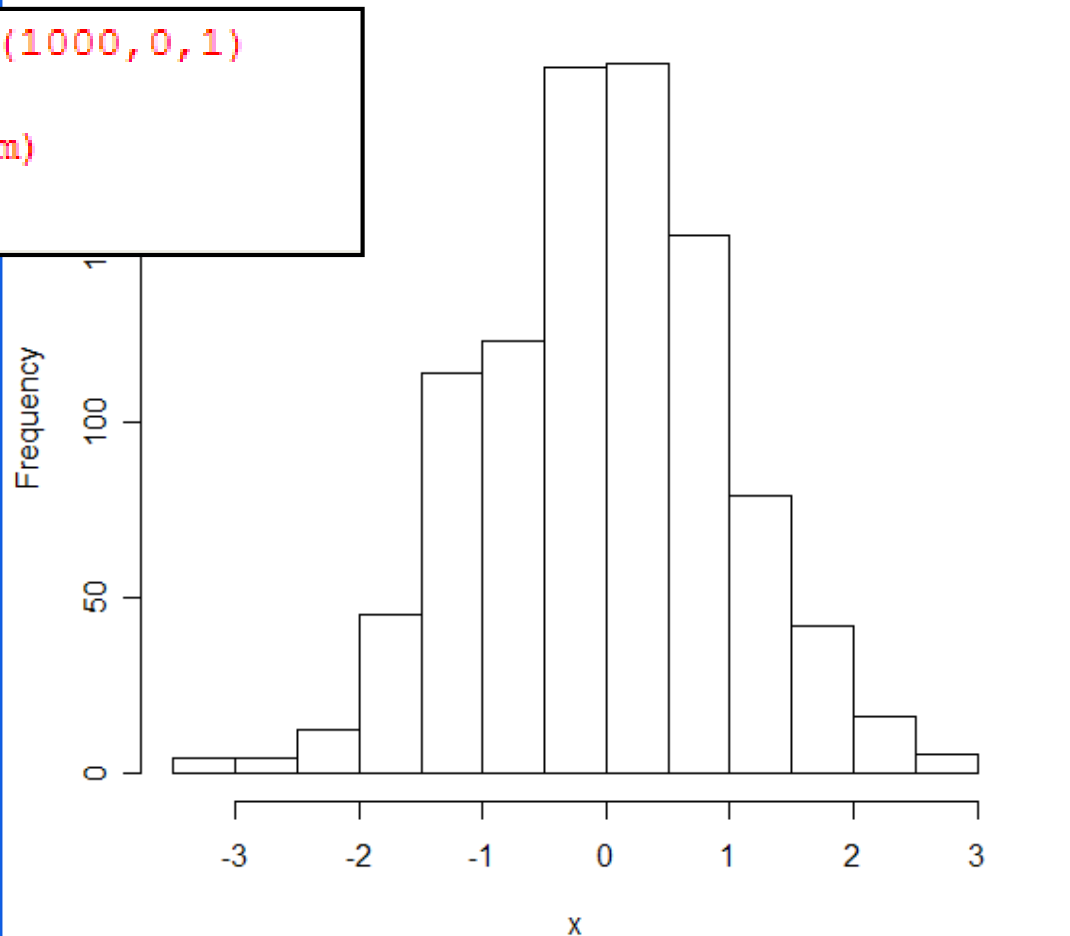
- Vector
- Matrix (二維)
- Array
- List
- Data Frame
- Function

R當計算機





Histogram of x



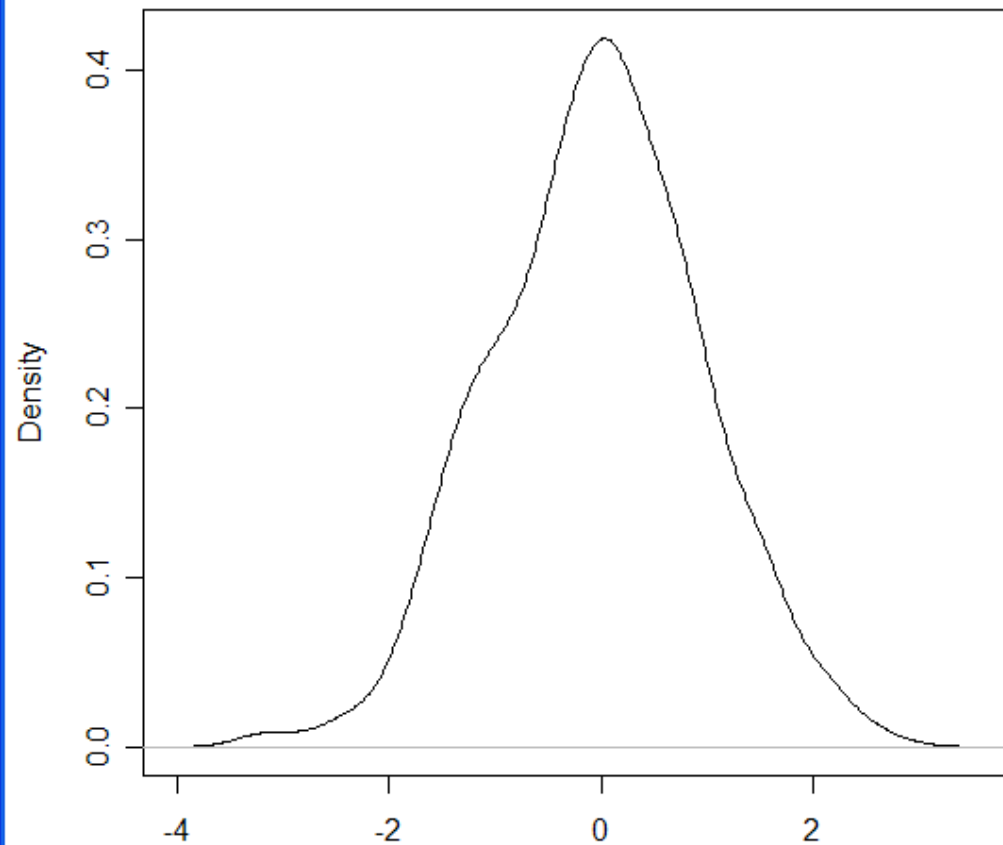
```
> x <- rnorm(1000,0,1)
> hist(x)
> help(rnorm)
> |
```

```
> x <-rnorm(1000,0,1)
> hist(x)
> help(rnorm)
> |
```

R Graphics: Device 2 (ACTIVE)



density.default(x = x)

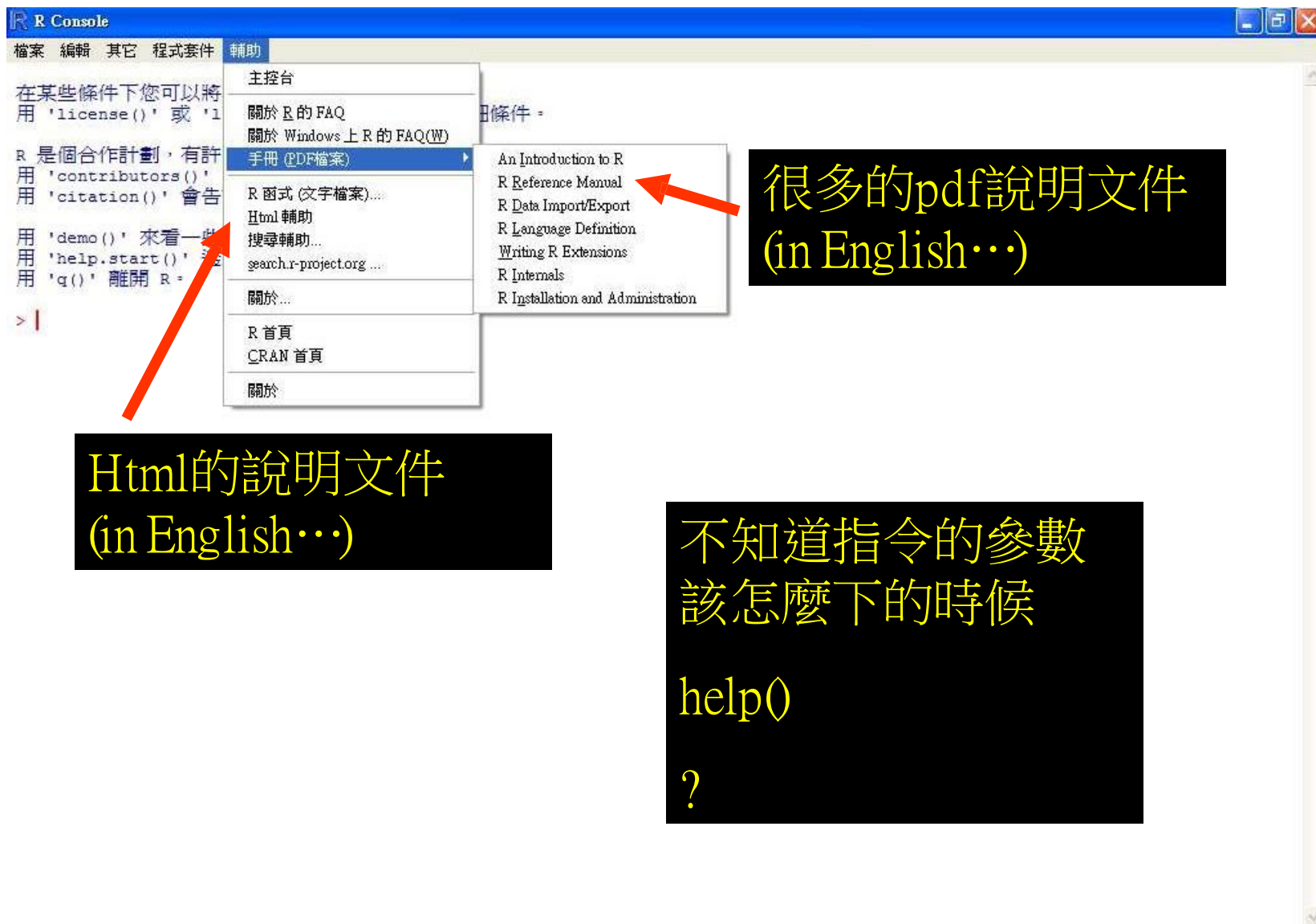


N = 1000 Bandwidth = 0.224

```
> plot(density(x))  
> █
```

設定工作目錄

- `getwd()` 檢視目前的工作目錄
- `#` 先開好資料夾 `C:\Rwork`
- `> setwd("C://Rwork ")`
- `# set new working directory`



安裝套件package

- `install.packages()`
- 使用GUI選單



選擇程式套件

```
R R Console

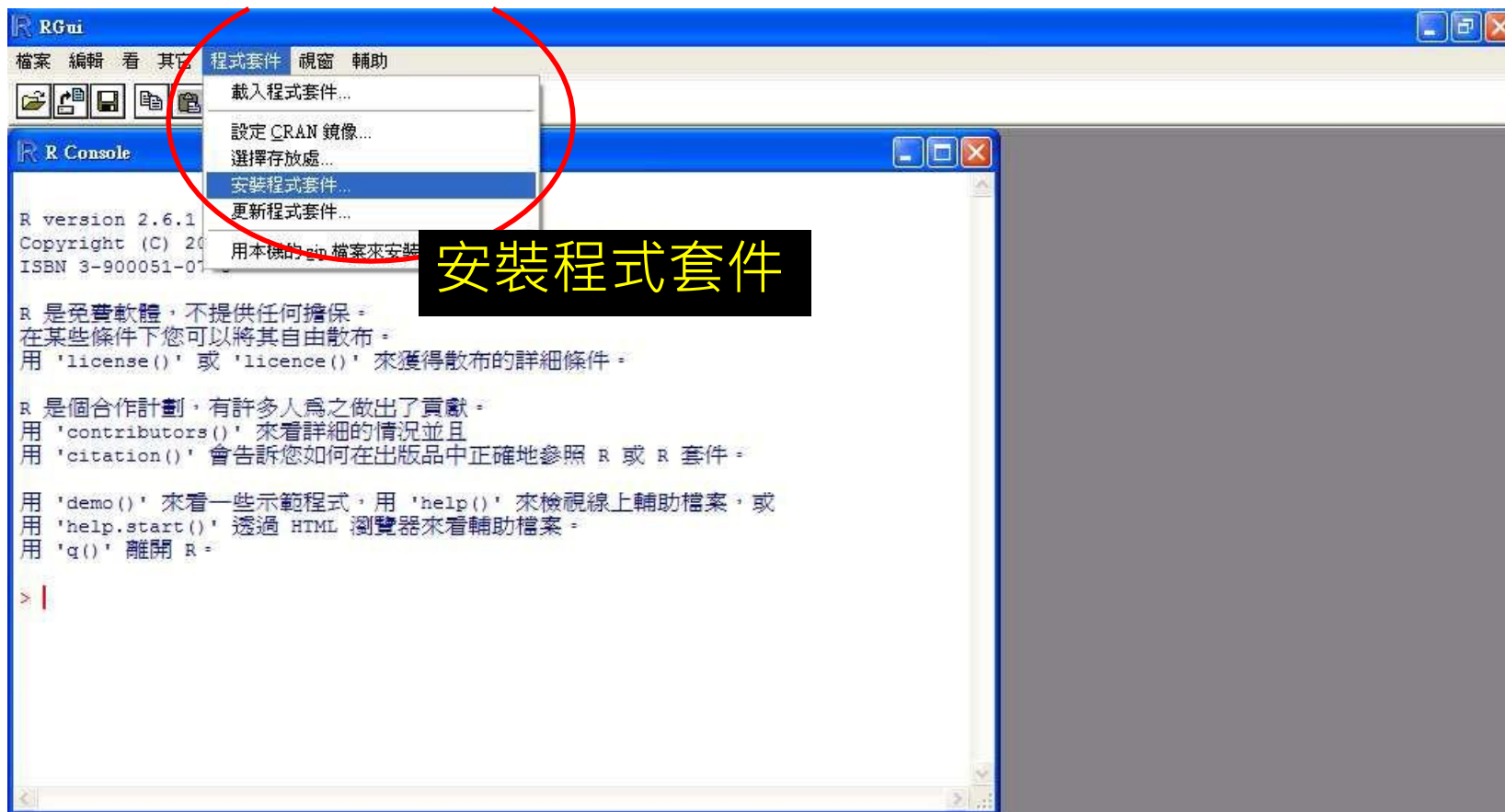
R version 2.6.1 (2007-11-26)
Copyright (C) 2007 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

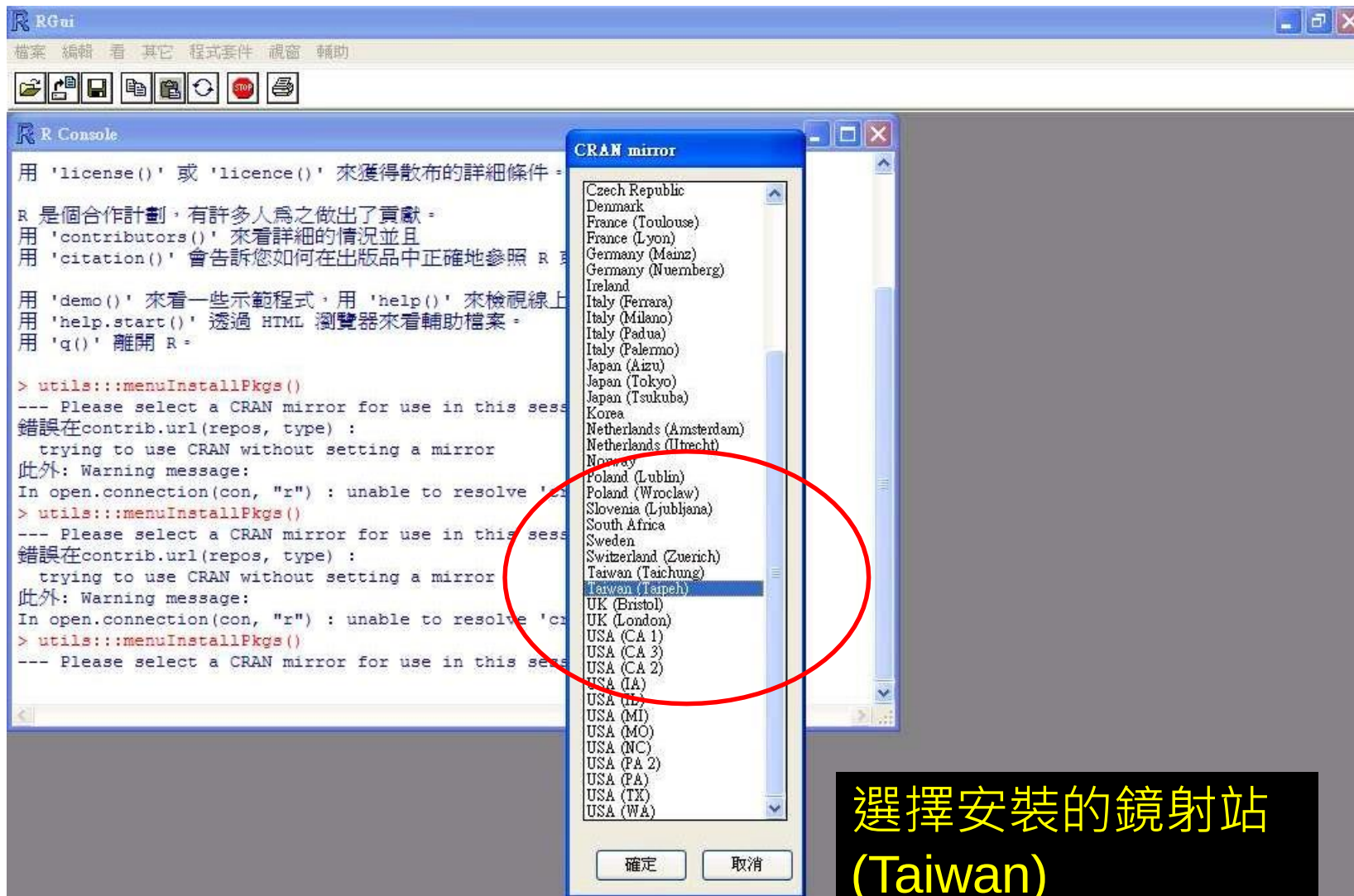
R 是免費軟體，不提供任何擔保。
在某些條件下您可以將其自由散布。
用 'license()' 或 'licence()' 來獲得散布的詳細條件。

R 是個合作計劃，有許多人為之做出了貢獻。
用 'contributors()' 來看詳細的情況並且
用 'citation()' 會告訴您如何在出版品中正確地參照 R 或 R 套件。

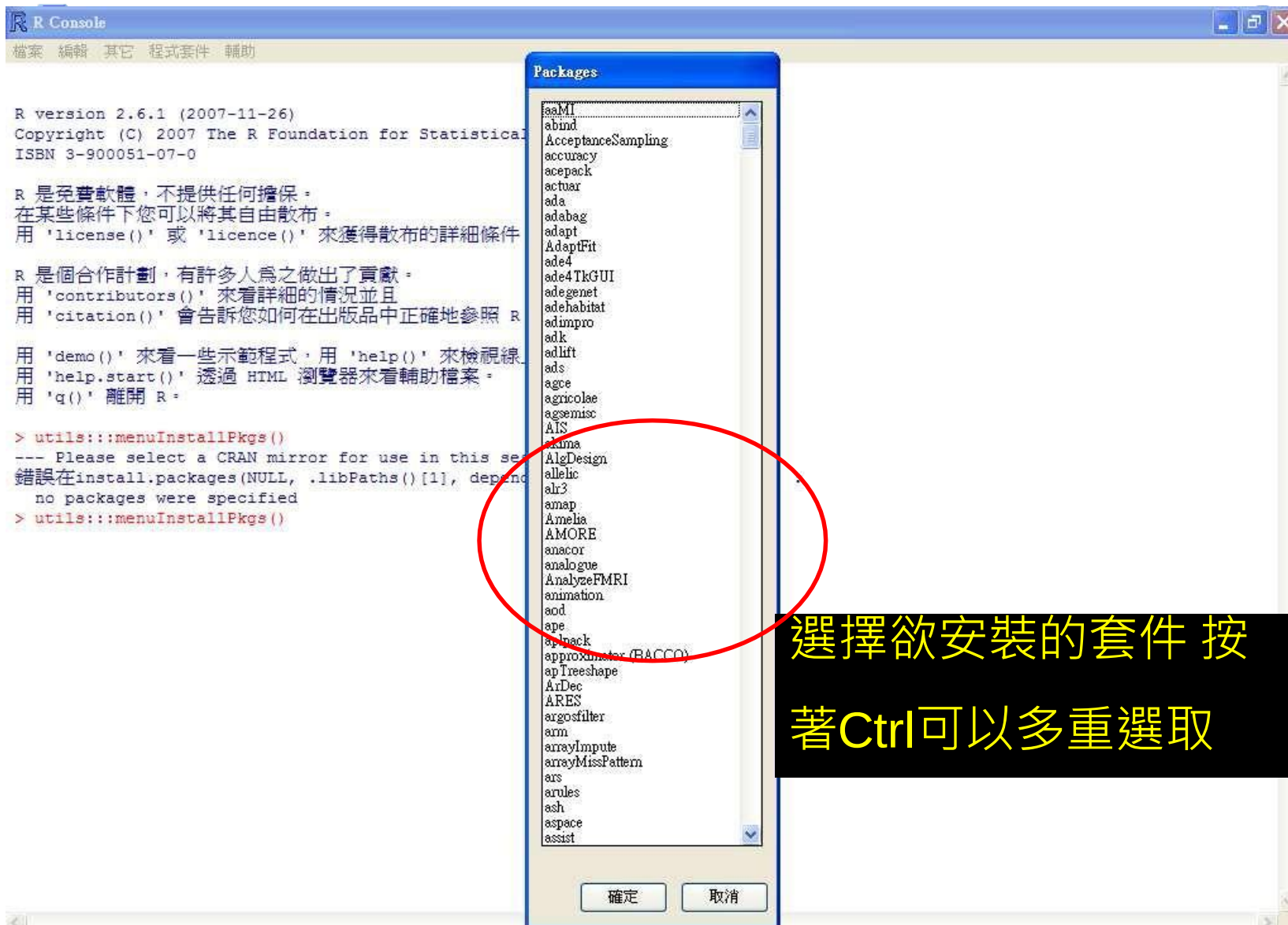
用 'demo()' 來看一些示範程式，用 'help()' 來檢視線上輔助檔案，或
用 'help.start()' 透過 HTML 瀏覽器來看輔助檔案。
用 'q()' 離開 R。

> |
```





選擇安裝的鏡射站
(Taiwan)



R version 2.6.1 (2007-11-26)

Copyright (C) 2007 The R Foundation for Statistical Computing

ISBN 3-900051-07-0

R 是免費軟體，不提供任何擔保。

在某些條件下您可以將其自由散布。

用 'license()' 或 'licence()' 來獲得散布的詳細條件。

R 是個合作計劃，有許多人為之做出了貢獻。

用 'contributors()' 來看詳細的情況並且

用 'citation()' 會告訴您如何在出版品中正確地參照 R 或 R 套件。

用 'demo()' 來看一些示範程式，用 'help()' 來檢視線上輔助檔案，或

用 'help.start()' 透過 HTML 瀏覽器來看輔助檔案。

用 'q()' 離開 R。

```
> utils:::menuInstallPkgs()
```

```
--- Please select a CRAN mirror for use in this session ---
```

```
錯誤在install.packages(NULL, .libPaths()[1], dependencies = NA, type = type) :
```

```
no packages were specified
```

```
> utils:::menuInstallPkgs()
```

```
試開URL 'http://cran.csie.ntu.edu.tw/bin/windows/contrib/2.6/zoo_1.4-2.zip'
```

```
Content type 'application/zip' length 932540 bytes (910 Kb)
```

```
開啓了 URL
```

```
downloaded 910 Kb
```

```
package 'zoo' successfully unpacked and MD5 sums checked
```

```
The downloaded packages are in
```

```
C:\Documents and Settings\person\Local Settings\Temp\RtmpYi2D0\downloaded_packages
```

```
updating HTML package descriptions
```

```
> |
```

完成套件的安裝

載入套件packages

- `library()`
- 使用GUI工具載入套件

Demo

- `demo()`
- `demo(package
= .packages(all.available = TRUE))`
- `demo(Japanese)`
- `demo(graphics)`
- `demo(image)`
- `demo(plotmath)`

Demo

- `library(tcltk)`
 - `demo(tkcanvas)`
 - `demo(tkdensity)`
- `library(rgl)`
 - `demo(rgl)`
- `library(seewave)`
 - `demo(seewave)`

R的相關網站

- R官方網站：<http://www.r-project.org>
- R繪圖：
 - <http://addictedtor.free.fr/graphiques/>
 - <http://cged.genes.nig.ac.jp/RGM2/index.php?clear=all>
- 中文教學網站：
 - 台北大學林建甫老師
<http://web.ntpu.edu.tw/~cflin/Teach/R/Rproj.htm>
 - 中山大學劉正山老師
<http://www2.nsysu.edu.tw/politics/liu/main/ResearchTools/appliedStatistics.htm>